MODEL QUESTION PAPER

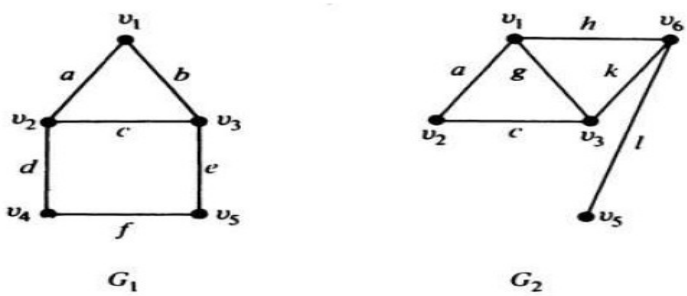
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: GAMAT401

Course Name: Mathematics for Computer and Information Science – 4

Max. Marks: 60		Duration: 2 hours 30 minutes

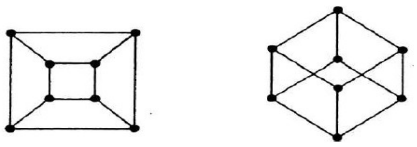
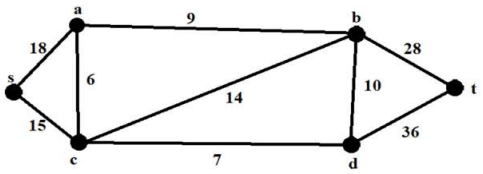
PART A

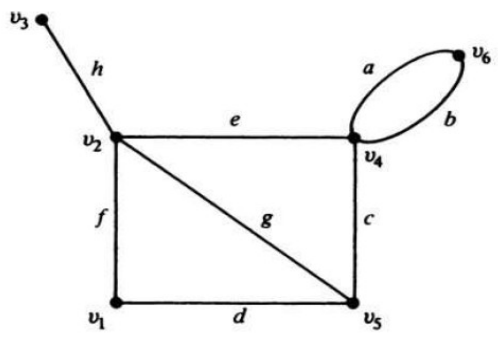
		Answer all questions. Each question carries (3) marks	CO	Marks
1		Prove that sum of the degrees of all vertices in G is twice the number of edges in G.	CO1	(3)
2		Define walk, path and circuit with examples.	CO1	(3)
3		Find the union of $G_1 and\ G_2$ and ring sum of $G_1\ and\ G_2$ 	CO2	(3)
4		Define Euler graph. Give an example of Eulerian graph which is not Hamiltonian.	CO2	(3)
5		Draw all unlabelled trees with 5 vertices.	CO3	(3)
6		Find the maximum number of vertices possible in a 3-level binary tree. Also find the maximum height possible in a binary tree with 11 vertices.	CO3	(3)
7		Show that every cut set in a connected graph must contain at least one branch of every spanning tree of G.	CO4	(3)
8		Prove that edge connectivity of a graph cannot exceed the degree of the vertex with the smallest degree in G.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries (9) marks

Module 1

9	a)	 Are the two graphs isomorphic ? Justify your answer.	CO1	(3)
	b)	Show that a simple graph with n vertices must be connected if it has more than $\frac{(n-1)(n-2)}{2}$ edges.	CO1	(6)
10	a)	Define complete graph and complete bipartite graph. Draw a graph which is a complete graph as well as a complete bipartite graph.	CO1	(4)
	b)	If a graph has exactly two vertices of odd degree, then prove that there must be a path joining these two vertices.	CO1	(5)
		Module 2		
11	a)	Define directed graphs. Differentiate between symmetric digraphs and asymmetric digraphs	CO2	(3)
	b)	Nine members of a new club meet each day for lunch at a round table. They decide to sit such that every member has different neighbours at each lunch. How many days can this arrangement last? Justify your answer.	CO2	(6)
12	a)	Define Hamiltonian circuit and Hamiltonian path. Give an example for each.	CO2	(3)
	b)	Prove that a connected graph is Euler if and only if it can be decomposed into circuits.	CO2	(6)
		Module 3		
13		Write down Dijkstra's algorithm and use it to find the shortest path from s to t.  	CO3	(9)
14		Show that a Tree with n vertices has exactly n-1 edges.	CO3	(9)
		Module 4		
15		Define cut-set. Prove that every circuit in G has an even number of edges in common with any cut-set.	CO4	(9)

16		For the following graph, find the (i) Incidence matrix (ii) Adjacency matrix (iii) Path matrix between $v_1 and\ v_4$ (iv) Circuit matrix  	CO4	(9)
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